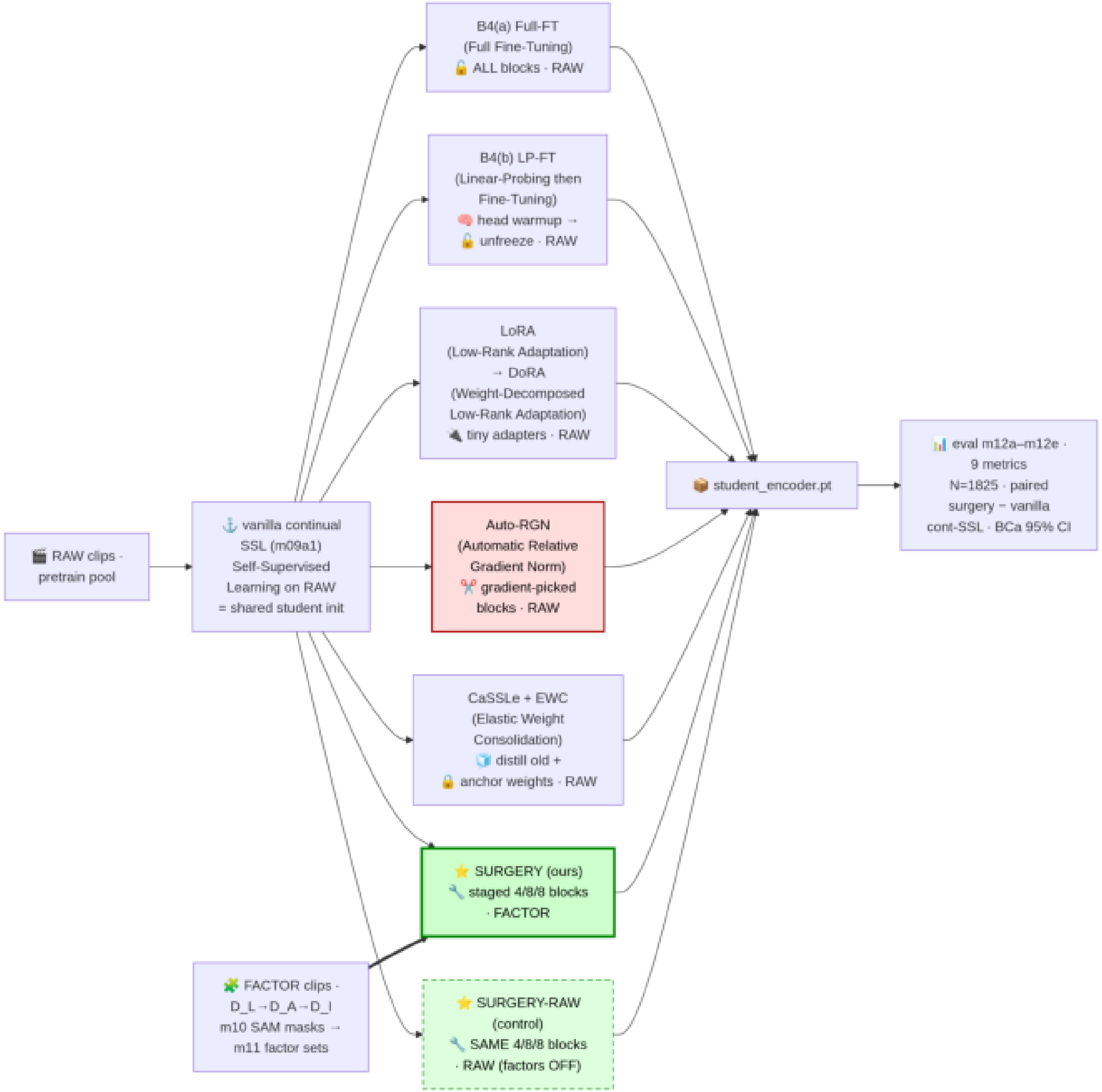
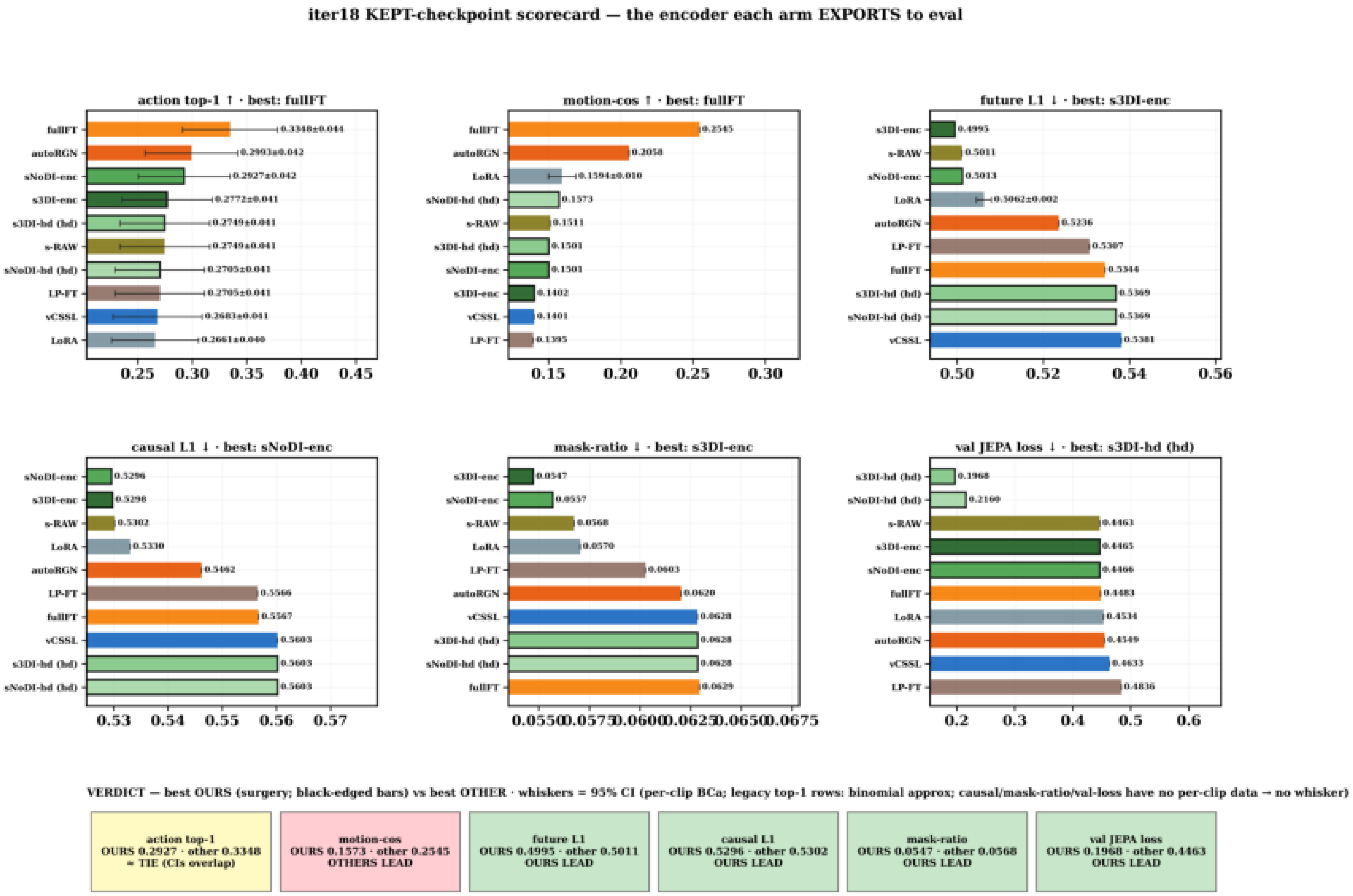
0 · Experiment map — 13 arms, one backbone (ViT-G 2B)



0 · Techniques — FULL FORM + like you're 5

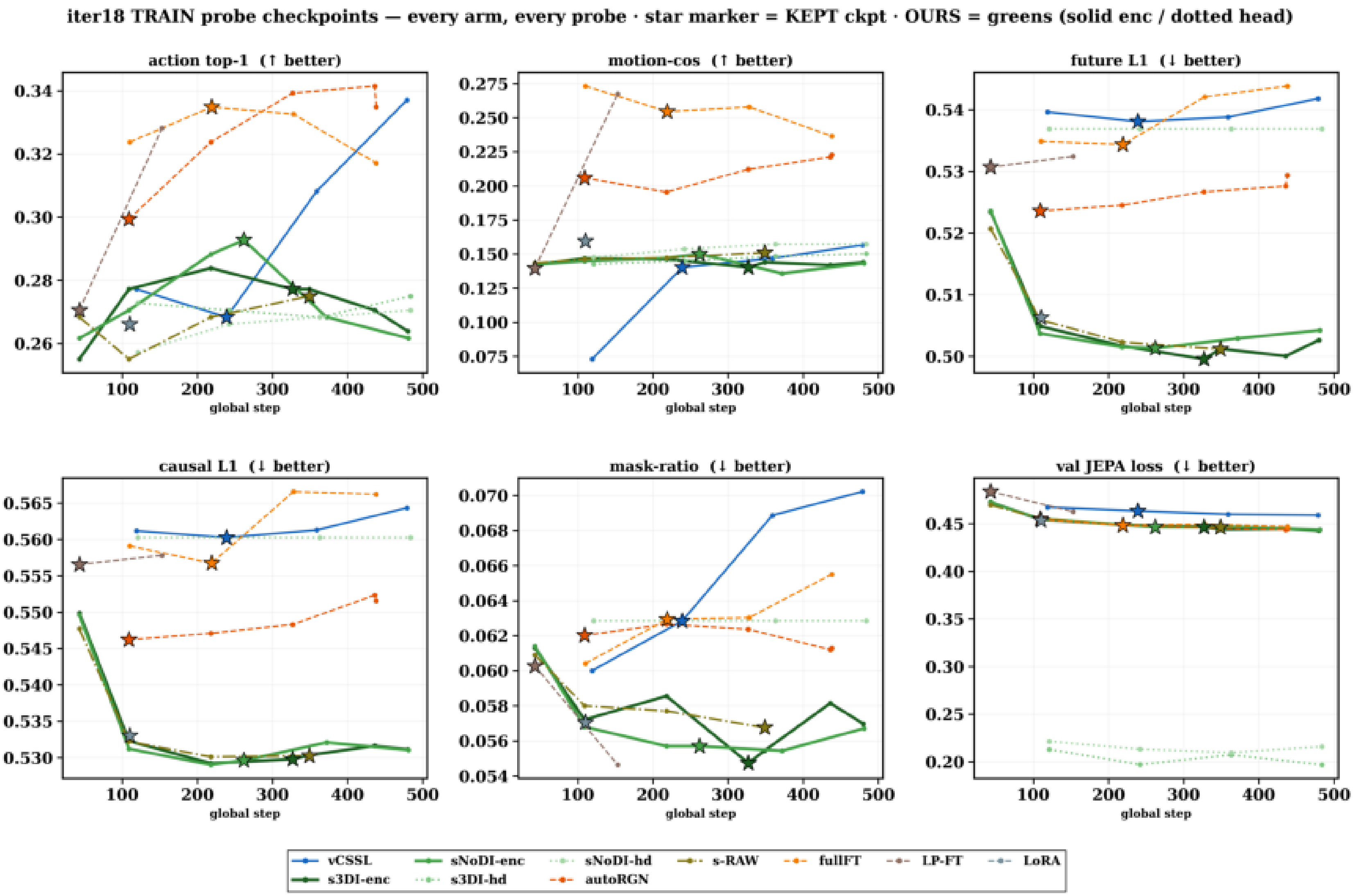
technique	FULL FORM (glossary)	like you're 5
frozen	frozen backbone (no adaptation)	Don't touch the brain; just test what it knows.
pretrain (vCSSL)	vanilla continual SSL (Self-Supervised Learning) — OURS anchor	Keep practicing videos the same old way.
surgery 3stage-DI (OURS)	staged factor-curriculum continual-FT, with D_I tubes	Special lesson clips; unlock slowly; includes things-touching clips.
surgery noDI (OURS)	staged factor-curriculum continual-FT, without D_I	Same slow unlocking; skip the things-touching clips.
surgery heads (OURS)	staged factor-curriculum continual-FT, head-only	Brain stays locked; only a tiny helper learns.
surgery RAW (control)	staged factor-curriculum continual-FT on raw clips	Surgery's slow unlocking, but with plain videos.
Auto-RGN	Automatic Relative Gradient Norm (Lee et al. ICLR'23)	Layers that pull harder get bigger learning steps.
Full-FT	Full Fine-Tuning	Unlock the whole brain; change everything at once.
LP-FT	Linear-Probing then Fine-Tuning (Kumar et al. ICLR'22)	First teach the helper, then unlock everything.
LoRA	Low-Rank Adaptation (Hu et al. 2021)	Stick tiny add-on notes; never rewrite the book.
DoRA	Weight-Decomposed Low-Rank Adaptation (Liu et al. 2024)	Sticky notes plus a volume knob per page.
CaSSLe	continual self-supervised distillation (Fini et al. CVPR'22)	New learning must still match the old teacher.
EWC	Elastic Weight Consolidation (Kirkpatrick et al. PNAS'17)	Important old memories get glued — hard to change.

1 · HERO — kept-checkpoint scorecard



- Surgery sweeps all three prediction metrics; full-FT sweeps semantics (top-1, motion-cos).
- Semantic gaps sit inside N=451 noise; surgery's prediction lead is consistent across variants.

2 · Training — every probe checkpoint, every arm



- Only surgery improves prediction metrics stage-by-stage; pretrain stays flat — progressive unfreeze works.
- Selector keeps each arm's future-L1 minimum, not its last checkpoint — mid-training peaks matter.

3 · Upcoming — per-encoder TEST eval scorecard

no EVAL artifacts yet —
per-encoder evals launch as each arm finishes training

- TEST verdicts (n=1,825, 95% BCa CIs) auto-fill as evals finish; val trends need confirmation.
- Paired-delta finale decides significance — val-probe leads are directional only.

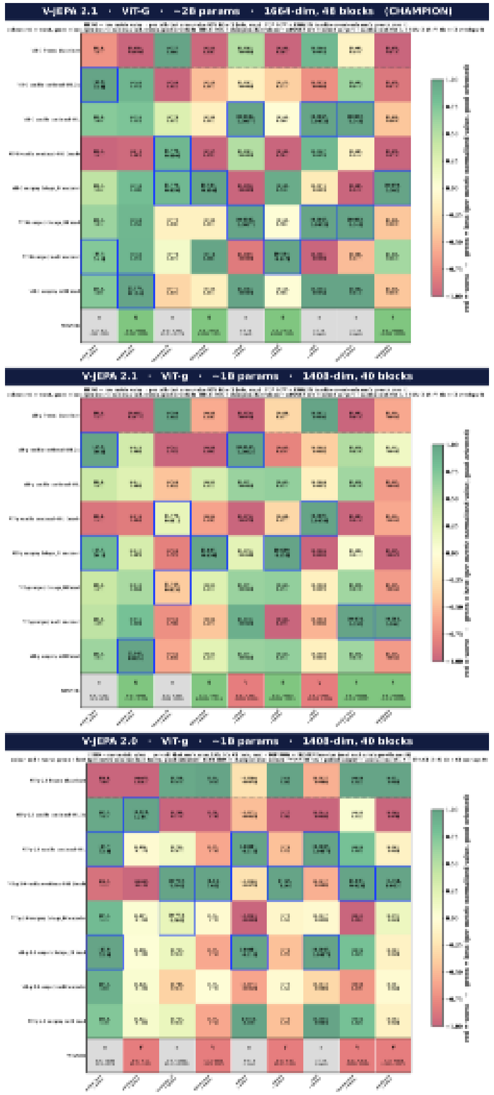
4a · iter17 — frozen backbone selection

FROZEN encoders — ABSOLUTE head-metric values (no Δ) · value ± BCa 95% CI · colour = per-metric min-max (green = best) · BLUE BOX = best per metric
10 frozen encoders × 3 shared metrics · sorted by action_top1 · 10 K bootstrap

	action_top1 ↑ better	motion_cos ↑ better	taxonomy_f1 ↑ better
V-JEPA 2.1 frozen	44.384 ±2.274	0.009 ±0.001	0.793 ±0.016
vjepa 2 1 vitL frozen	44.219 ±2.301	0.004 ±0.000	0.788 ±0.017
vjepa 1 vitH frozen	40.493 ±2.219	0.007 ±0.001	0.702 ±0.019
lejepa vitL frozen	40.055 ±2.219	0.014 ±0.002	0.740 ±0.018
vjepa 1 vitL frozen	39.890 ±2.274	0.008 ±0.001	0.660 ±0.019
ijepa vitH14	39.123 ±2.247	0.016 ±0.001	0.781 ±0.017
vjepa 2 0 vitg ssv2	38.849 ±2.274	0.007 ±0.001	0.776 ±0.017
DINOv2 frozen	38.466 ±2.247	0.016 ±0.001	0.816 ±0.016
vjepa 2 vitL 256 frozen	37.918 ±2.247	0.013 ±0.001	0.778 ±0.017
ijepa vitG16	37.480 ±2.219	0.019 ±0.001	0.787 ±0.017

- V-JEPA 2.1 ViT-G is the strongest frozen backbone — justified as iter18's sole backbone.
- Every frozen model is motion-blind (motion-cos ~ 0) — adaptation is necessary, not optional.

4b · iter17 — ViT-G family raw values after training



- Any continual training beats frozen by +6-7pp top-1; gains are not surgery-specific.
- Only surgery moves future-MSE and causal — encoder updates drive temporal gains, heads don't.

4c · iter17 — paired deltas with 95% CI

PAIRED surgery − vanilla cont-SSL (champion duel) across backbones · per cell: Δ + 95% BCa CI
deep green = surgery sig. better · deep red = vanilla cont-SSL · pale grey = TIE (95% CI crosses 0) · 10 K bootstrap

	V-JEPA 2.1 ViT-G ~2B params	V-JEPA 2.1 ViT-g ~1B params	V-JEPA 2.0 ViT-g ~1B params
action_top1	-1.37 [-3.12, +0.329] TIE	-0.438 [-2.30, +1.48] TIE	+0.219 [-1.26, +1.81] TIE
motion_cos	+0.010 [+0.0085, +0.012] SURGERY	+0.018 [+0.017, +0.020] SURGERY	-0.102 [-0.105, -0.099] VANILLA CONT-SSL
taxonomy_f1	-0.0001 [-0.015, +0.015] TIE	-0.0085 [-0.024, +0.0066] TIE	-0.0080 [-0.025, +0.0089] TIE
future_mse	+0.025 [+0.024, +0.025] SURGERY	+0.0037 [+0.0030, +0.0044] SURGERY	-0.070 [-0.072, -0.067] VANILLA CONT-SSL
rollout	0 ≈ equal TIE	-0.0001 [-0.0002, -0.0000] VANILLA CONT-SSL	0 ≈ equal TIE
causal	+0.019 [+0.018, +0.020] SURGERY	+0.0012 [+0.0008, +0.0015] SURGERY	-0.062 [-0.065, -0.060] VANILLA CONT-SSL
tdist	0 ≈ equal TIE	-0.0009 [-0.0011, -0.0008] VANILLA CONT-SSL	0 ≈ equal TIE
teacher_free	0 ≈ equal TIE	+0.0014 [+0.0011, +0.0017] SURGERY	-0.0019 [-0.0022, -0.0016] VANILLA CONT-SSL
maskratio	+0.0091 [+0.0088, +0.0094] SURGERY	+0.0018 [+0.0016, +0.0021] SURGERY	-0.0078 [-0.0085, -0.0070] VANILLA CONT-SSL
WINS	Surgery 4 Vanilla cont-SSL 0 tie 5	Surgery 5 Vanilla cont-SSL 2 tie 2	Surgery 0 Vanilla cont-SSL 5 tie 4

- Surgery beats frozen significantly on action and motion-cos — CIs exclude zero.
- Versus pretrain: action ties; motion/prediction edge persists — surgery's value is temporal, not semantic.